Final Project

Intelligent Robots: Algorithms and Systems



Observation-Driven Grasping and Object-Aware Maze Traversal

CS4278 / CS5478 Intelligent Robots

Final Project Report

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Submitted by

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Observation-Driven Grasping and Object-Aware Maze Traversal

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Abstract

We present a complete controller for the provided Fetch pick–navigate–reach task. From object points, surface normals, robot state, observed maze walls and the goal, it selects a shape-aware grasp, executes collision-checked arm motion, chooses a corridor-compatible carry pose and routes the combined robot–object footprint. On 15 provided objects and 100 seeds per object, the automatic controller lifted and held **1,433/1,500** (95.5%) and delivered **1,390/1,500** (92.7%) within the 10 cm goal tolerance. Of 110 end-to-end failures, 67 occurred during pickup and 43 after pickup, including 26 transport drops. Keeping finger closure commanded throughout top-down carry motions avoids a brief loss of grip pressure. On the same seeds, deliveries rose from 1,207/1,500 before this fix to 1,390/1,500 afterward. These are seeded evaluations of the supplied environment, not estimates for hidden objects or unseen maze layouts.

1 Problem and design choices

The course task asks a Fetch mobile manipulator to pick a provided YCB-derived object from a table, carry it through a wall maze and bring its center within 0.10 m of a three-dimensional goal [1, 5]. The simulator accepts absolute position targets for 15 controllable joints. Its observations include 1,024 object surface samples and outward normals, current joint state, object and goal poses, and the XY positions of wall cubes. The policy does not read a precomputed solution path or attach an object to the gripper.

Pickup success alone is not sufficient: a grip that lifts a bowl may peel off its rim, and a grip that holds a hammer at the table may lose it when the arm stows. We consequently evaluate the grasp after a 10 cm lift and a 50-step hold, then monitor the same physical hold throughout navigation. A valid delivery additionally requires a goal-distance check after a final 50-step hold. Figure 1 illustrates the geometry addressed by the controller.

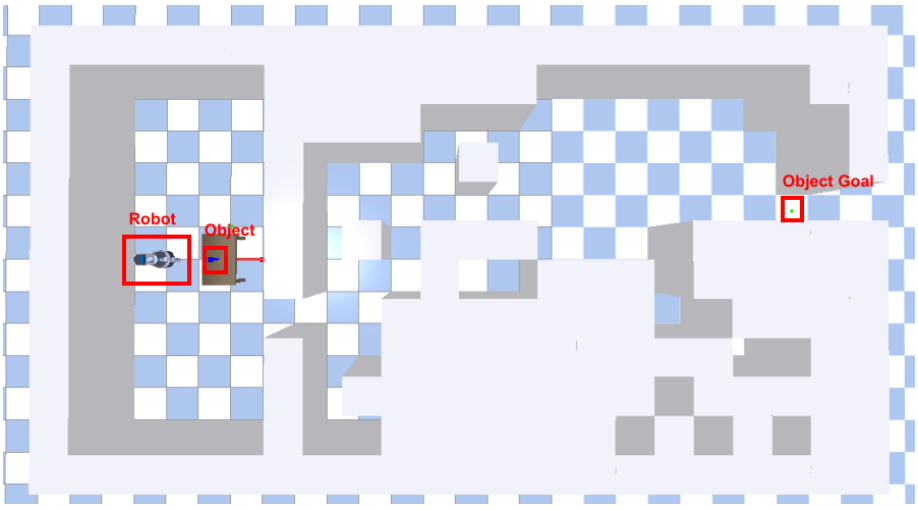


Figure 1: Provided scene: the robot must move a grasped object from the table through observed walls to the goal. The route is checked for both the robot and the held object's footprint.

Our solution has three coupled decisions: find contacts that fit the jaws and are reachable without hitting the table; move the hand to a carry pose that fits the maze; and route the base while accounting for the object's measured offset. This coupling matters especially for wide, rimmed and flat objects.

2 Observation-driven grasp selection

2.1 Contact proposals and executable arm motion

We deterministically subsample at most 256 of the 1,024 observed surface points, then propose nearly horizontal, opposing contact pairs. For two points p_i, p_j with outward normals n_i, n_j, let

$$u = \frac{p_j - p_i}{\|p_j - p_i\|}, \qquad (-n_i)\cdot u > 0.85, \quad n_j \cdot u > 0.85. \tag{1}$$

Pair height difference is below 18 mm and width is normally below 85 mm; the threshold can rise to 100 mm for observed wide objects. Candidate scores favor opposing alignment and penalize displacement from the observed center. After near-duplicate suppression, the planner tries at most 30 different grasps. This antipodal-style construction is motivated by classical grasp geometry and point-cloud grasp detection [2, 3]; the alignment test is a heuristic, *not* a proof of force closure.

For each candidate, the controller solves wrist pose and arm joint targets in an isolated PyBullet kinematics client. It checks the planned joint segments at intermediate configurations against the table and rejects an approach whose gripper palm contacts the object before its fingers can close. The live robot only receives joint-position commands; table clearance, joint motion and finger proximity are checked again during execution. We approach from above by default and use a near-flank side pinch for geometries such as the wide Master Chef can. A long, flat object with a narrow handle is tried from above first: the hammer's broad head would otherwise trigger a side approach that lifts but can rotate free during transport.

2.2 Rim and plate are distinct geometries

Ordinary opposing sidewalls do not describe a thin bowl rim. For a bowl-like point cloud, we seek nearby inner/outer rim samples with opposing local normals and support on both walls. The gripper approaches this radial pair from above, matches its closure to rim thickness and lifts in short, straight-up Cartesian IK segments. A long joint-space interpolation could sweep the wrist laterally and peel the rim out of the jaws.

A plate lying flush with the table has no clearance for a lower finger. After direct plans fail, a geometry-triggered fallback pushes it toward the observed table edge with closed fingers, obtains a fresh point cloud and pinches the newly exposed lip with one finger above and one below. The plate is never teleported or rigidly attached. Figure 2 depicts the result of the staged pickup.

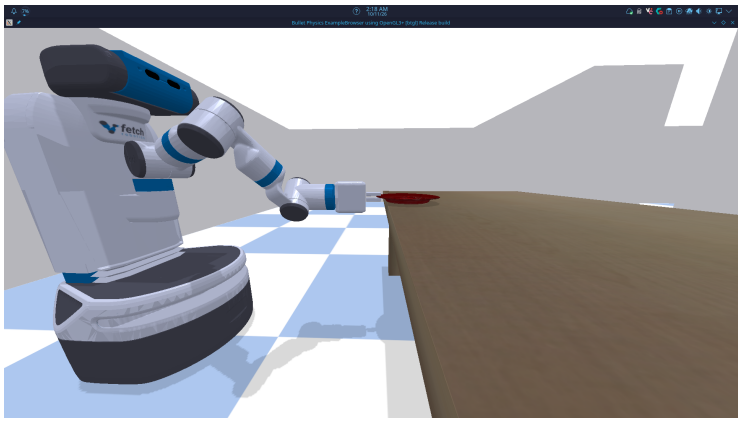


Figure 2: The physically staged plate exposes an edge for an upper/lower side pinch. Its initial flush pose cannot be gripped this way.

3 Carrying, mapping and reaching

After a measured lift, the controller may roll a top-down grasp in safe 15-degree increments to give one jaw better support; long thin objects are limited to one such increment. It searches IK-valid stow poses with the held object close to the base. Each candidate is accepted only if its projected robot–object footprint admits a collision-checked maze route. Thus the carry pose is chosen *before* committing to the wall-grid plan; a successful table pickup cannot by itself guarantee a traversable carried configuration.

While an object is pinched, the measured finger joints remain partly open even though their commanded target is closed. Interpolating from that measured opening during the post-lift arm roll and stow briefly relaxes the grip. For top-down grasps we instead hold the existing closed finger target through these arm motions. Finger-force limits and collision safeguards are unchanged; specialized side/plate grasps retain their separate stow behavior.

We construct a two-dimensional grid from the observed wall positions. Let b denote base XY, δ the measured object-center offset from the base, B the robot's observed XY collision bounds and O the carried object's observed XY half-extents. A base location is admissible only if the translated base bounds $b + B$ and object bounds $b + \delta + O$ both clear the walls (with a small margin). A* searches cardinal neighboring free cells and merges only collinear, collision-checked waypoints [4]. Connectors to the grid and goal use checked orthogonal legs; the base commands one translation axis at a time rather than cutting diagonally

across wall corners. The nominal goal-base position is $b^* = g_{xy} - \delta$; it may be shifted slightly within the goal tolerance when the corridor exit requires clearance. During motion we check live robot and object wall distances as well as the object–hand displacement.

At the exit, the arm adjusts the held object's height and the base trims small XY errors before the final goal and hold check. A staged plate can leave the arm folded near several joint limits: if no ordinary collision-free final lowering pose exists, a plate-only fallback first verifies a 0.5 m retreat through open space, then extends and lowers the grasp while preserving jaw closure. This resolves a reachability problem without requiring a wall collision or an object release.

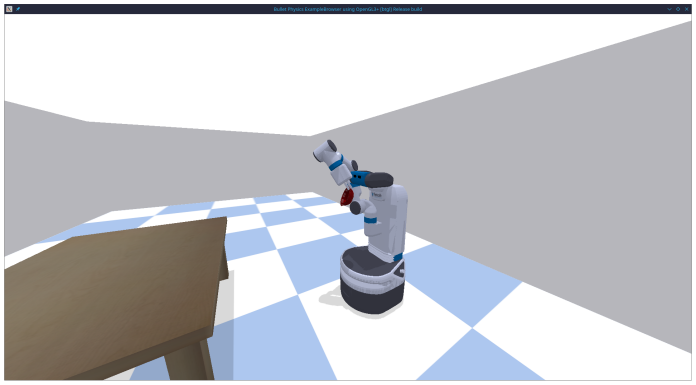


Figure 3: A measured carry pose stows the held object close to the base before the robot enters the wall grid; the grid subsequently protects both.

4 Evaluation and results

4.1 Protocol and aggregate outcomes

We ran independent headless full-task trials for all 15 supplied object types and seeds 100–199: **1,500 attempts**. Pickup and delivery counts below come from these *same* trials, not from separate pickup-only runs. Each run starts from a fresh simulator, settles the object for 80 environment steps, then executes the automatic policy. The maze wall generator uses a fixed internal seed; trial seeds change sampled object surfaces and randomized entry/exit rows. This seeded evaluation is not a hidden-maze test. A pickup must rise at least 0.10 m and survive 50 held steps; end-to-end success must also pass the simulator's 3D goal test after transport. An environment step can contain several Bullet physics ticks; reported timings below count environment steps.

Table 1: Successful pickups and full deliveries per object (100 seeds each). Bowl-rim grasps are counted under the benchmark's top-down label.

Object	Pick	Goal	Object	Pick	Goal
Chef can	85	71	Bleach	99	99
Sugar box	100	94	Bowl	87	87
Soup can	100	100	Mug	98	97
Mustard	83	83	Plate	88	66
Potted meat	100	100	Large marker	100	100
Banana	100	100	Hammer	93	93
Strawberry	100	100	Foam brick	100	100
Apple	100	100			
Total across 1,500 trials				**1,433**	**1,390**

The controller picked up **1,433/1,500 (95.5%)** and delivered **1,390/1,500 (92.7%)**; delivery conditional on pickup was 1,390/1,433 = 97.0%. Successful pickups comprise 1,173 ordinary top-down, 87 bowl-rim top-down and 173 side grasps (85 can, 88 staged plate). The benchmark's coarser labels are 1,260 top-down and 173 side. Among 67 pickup failures, 23 objects did not lift, 15 approaches stopped at the table-clearance guard, 14 missed a safe waypoint, ten found no safe grasp approach and five had other guarded or tracking failures. Among 43 post-pickup failures, 26 were transport drops. Table 1 gives the per-object results. Median first-lift time among successful pickups was 1,613 environment steps; median

navigation time among deliveries was 4,009.5 steps. The longest successful navigation took 5,138 steps, below the 15,000-step limit.

4.2 Post-routing grip-retention improvement

In a saved run on the *same* seeds 100–199 after cardinal routing was in place but before the finger-command fix, 1,433/1,500 pickups led to 1,207/1,500 deliveries (80.5%); 209 objects dropped during transport. Holding the closed finger target during top-down support rolls and stowing left pickups at 1,433/1,500, raised deliveries to 1,390/1,500 (92.7%, a gain of 12.2 percentage points) and reduced drops to 26. In paired trials, 185 changed from failure to delivery and two from delivery to failure: a net gain of 183 deliveries. This improvement is due to the later *grip-command* change, not the earlier diagonal-route removal. The pre-fix data are preserved in Git revision `666f3cd:full_traversal_results.json`; the post-fix data are the current root JSON. Neither is pooled with the development results below.

4.3 Earlier development and carry ablation

On *historical development seeds 60–69*, an early policy held 126/150 objects; local bowl-rim grasps raised this to 135/150, and staged plate pickup reached 142/150 in an intermediate version. A later development version held 145/150 and delivered 135/150. These are engineering history on ten seeds per object, *not* results to pool with the present seeds 100–199. In a separate, earlier *paired* carry-pose ablation on 150 development trials, a full 180-degree palm-up flip delivered 87/150 versus 128/150 for the then-default small jaw-roll policy; both picked up 142 objects. The full flip often had no collision-free pose and was rejected. Neither comparison tests the current grip-retention fix on the present 1,500-trial cohort.

4.4 Diagnosing the remaining failures

The 26 transport drops comprise ten Master Chef cans, nine plates, six sugar boxes and one mug. The live monitor marks a lost grasp when the object deviates more than 0.12 m from its hand-relative position or falls more than 0.12 m below its carried height. These are observations of failed retention, not evidence that A* selected an unsafe maze route. No trial stopped at the wall-proximity guard; two post-pickup trials found no collision-free maze route or carry pose. The plate held in 88/100 trials but delivered in 66/100; besides nine drops, its post-pickup failures include seven final-reach pose failures, five unstable-joint stops and one goal-tolerance miss. Seven delivered plate trials used the guarded open-space retreat. The can held in 85/100 but delivered in 71/100; 13 of its pickup failures hit the table-clearance guard.

End-to-end success *given pickup* is $1{,}390/1{,}433 = 97.0\%$. This rate does not remove the plate and can bottlenecks: a grip can survive the lift but fail while stowing, traversing or reaching the goal.

4.5 Interpretation of the policy iterations

Table 2 records only the pickup-stage *development* sequence. The counts use the same 60–69 seeds, but the controllers changed between rows; they are not independently randomized trials or confidence intervals. The first gains came from treating a bowl rim as *local* opposing walls and physically exposing a plate lip before pinching it. The subsequent hammer handle preference improved retention without changing the number of objects initially lifted in that small cohort.

Table 2: Historical pickup-policy development on seeds 60–69 (150 trials per row); these are *not* the current 1,500-trial results.

Automatic pickup policy	Held after lift + 50 steps
Initial antipodal and IK policy	126/150
Add inner/outer bowl-rim lift	135/150
Add staged plate-edge pickup (intermediate)	142/150
Later geometry-specific pickup and grasp selection	145/150

The aggregate lift-time median also conceals a practical tradeoff. Most successful lifts clear the 10 cm threshold relatively quickly, but plate trials use many short, checked motions for the physical slide, reobservation and edge pinch. Their median successful first lift is 5,576 environment steps, well beyond the course's 3,000-step pickup target. The gripper is kept close to the table during this procedure, so simply issuing a larger or faster joint command would remove the safeguards that prevent fingertip and wrist

collisions. Reducing redundant staging movement while retaining measured clearance is a more defensible performance target than weakening collision checks. These timing observations are from simulator actions, not wall-clock seconds or a hardware Fetch robot.

The earlier full palm-up ablation tested a different pickup policy and older plate behavior, so it cannot identify a universal "best orientation". Its useful observation is narrower: a large flip often has no safe arm pose and can destabilize a hold. We do not combine its 87/150 and 128/150 deliveries with today's 1,390/1,500 result.

5 Limitations and conclusion

The remaining 110 unsuccessful full trials include 67 pickup failures and 43 post-pickup failures; 26 of the latter are transport drops. The plate's median successful lift takes 5,576 environment steps, above the course's 3,000-step pickup timing target, and only 66/100 plate trials deliver. The can delivers in 71/100 trials, often limited by table clearance or grip retention. The controller relies on sampled surface geometry, fixed-heading base translation, approximate footprint bounds and a fixed-maze generator; we have not established robustness to unseen object poses, obstacle layouts or sensor noise. Its contact checks are heuristics rather than a force-closure or global motion-planning guarantee.

The bounded grasp search still spends time on many alternative contact pairs and IK branches. Each geometric proposal is evaluated against a kinematic model, while the executed robot can lag commanded joint targets. This is why we check measured pose and clearance during motion rather than trusting collision-free IK endpoints alone. The finite sampled surface may also miss an exposed contact, and axis-aligned object bounds can overstate clearance needs for rotated shapes. Both choices favor safety on these provided scenes but can cause avoidable failures on different layouts.

The main result is an end-to-end lesson: selecting contacts, moving safely to a grasp, choosing a carry pose and routing the held object are inseparable. Rim-aware lifting and staged edge exposure address geometry at the table; object-aware A* and a guarded open-space final reach preserve delivery. Across the provided 100-seed cohort this yields 1,433/1,500 measured pickups and 1,390/1,500 full deliveries, with remaining work concentrated in can and plate reliability and the slow plate staging sequence.

Reproducibility

The current trial-level data file is:
`full_traversal_results.json` (repository root). Its per-trial `picked` and `success` fields generate both sets of counts. The file records seeds 100–199, 15 objects per seed, failure stages and environment-step timings. The two benchmark JSON files under `presentation/` are *historical* seeds 60–69, not inputs to the current figures. The controller and `benchmark_full_task.py` accompany this report; the 1,500-trial benchmark was not rerun to prepare these deliverables. No claim is made about the five unprovided course evaluation seeds.

References

[1] NUS-LinS-Lab. *Pick-Navigation-Reach Default Project*, CS4278/CS5478 course environment and rubric, project repository and supplied task brief. `https://github.com/NUS-LinS-Lab/Pick-Nav-Reach-Default-Project`.

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